Optimization II — Project 2

Dynamic Programming for Airline Revenue Management

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Problem Overview

We work for an airline's pricing department and want to determine the best ticket prices to charge each day over a 365-day booking horizon in order to maximize expected profit. The flight has two cabin classes: coach with 100 seats and first-class with 20 seats. Each day, the airline chooses a price for each class, and customers either buy a ticket or don't, based on the price. At the end of the 365 days, passengers show up (or don't), and if more passengers show up than there are seats, the airline pays penalty costs.

Revenue comes from selling tickets at the chosen daily price. Costs arise only when the flight is overbooked: if a coach passenger can be bumped up to an available first-class seat, the airline pays $50; if a passenger cannot get on the plane at all, the airline pays $425. First-class is never overbooked by policy, only coaches can be oversold.

The three possible coach prices are $300, $325, and $350. At $300 there is a 65% chance a coach ticket is sold on any given day; at $325 there is a 45% chance; and at $350 there is a 30% chance. For first-class, the two options are $425 (8% chance of a sale) and $500 (4% chance of a sale). If first-class has already sold out, coach demand gets a bonus of 4 percentage points at any price, since some first-class customers will choose coach instead.

Money earned in the future is worth slightly less than money earned today. We account for this using a daily discount factor derived from a 17% annual interest rate:

delta = 1 / (1 + 0.17 / 365) = approximately 0.99953 per day

This means a dollar earned tomorrow is worth about 0.99953 dollars in today's terms. Over 365 days, this compounding effect is meaningful.

Mathematical Formulation

State

At any point during the booking period, the state is described by three pieces of information:

- t = number of days remaining until departure (0 to 365)
- c = number of coach tickets sold so far
- f = number of first-class tickets sold so far

Value Function

The value function V(t, c, f) represents the maximum additional expected discounted profit we can earn from day t onward, given that c coach tickets and f first-class tickets have already been sold.

Bellman Equation

The Bellman equation states the best value at any state equals the best expected immediate reward plus the discounted best value in the next state.

V(t, c, f) = max over (price_coach, price_first) of:
E[Revenue_today + delta x V(t+1, c', f')]

That is, each day we try every combination of coach and first-class prices, compute the expected revenue plus the discounted future value, and pick the price pair that gives the highest total.

Discount Factor

A dollar earned today is worth slightly more than a dollar earned tomorrow. The daily discount factor reflects the 17% annual interest rate:

delta = 1 / (1 + 0.17 / 365) approximately 0.99953

Each future cash flow is multiplied by delta for each day it is delayed.

Demand Probabilities (State Transition)

On any given day, four outcomes can occur (assuming independence between coach and first-class demand):

Outcome	Probability
No tickets sold	(1 - p_coach) x (1 - p_first)
Only a coach ticket sold	p_coach x (1 - p_first)
Only a first-class ticket sold	(1 - p_coach) x p_first
Both a coach and a first-class ticket sold	p_coach x p_first

Where p_coach and p_first are the sale probabilities at the chosen prices for that day.

Terminal Condition: Overbooking Cost at Departure

On day t = 365 (departure day), there are no more ticket sales. Instead, we compute the expected penalty cost based on how many passengers actually show up.

The number of coach passengers who show up follows a Binomial distribution:

$$C_show \sim Binomial(c, 0.95)$$

The number of first-class passengers who show up follows:

$$F_show \sim Binomial(f, 0.97)$$

If coach show-ups exceed capacity:

$$Overflow = \max(C_show - 100, 0)$$

Some overflow passengers may be bumped up to first-class if seats are available there. The rest are denied boarding. The cost formula is:

$$Cost = (50 \times bumped_to_first) + (425 \times denied_boarding)$$

The expected terminal cost is calculated by summing over all possible show-up combinations, weighted by their binomial probabilities.

Task 1 - Expected Profit with 5 -seat Overbooking

Setup

For Task 1, we allow the airline to oversell coach by 5 seats. This means the airline can sell up to 105 coach tickets (100 seats + 5 extra). The dynamic program is solved over all states (t, c, f) where:

- t ranges from 0 to 365
- c ranges from 0 to 105 (max coach tickets sold)
- f ranges from 0 to 20 (max first-class tickets sold)

How the Algorithm Works

The algorithm starts at t = 365 and computes the expected overbooking cost for every possible (c, f) combination. It then works backward day by day. At each day t and state (c, f), it evaluates all six price combinations and selects the one that maximizes the sum of expected daily revenue and the discounted value of the next state. This process continues all the way back to t = 0.

The key insight is that later decisions (closer to departure) affect earlier ones. Selling too many coach tickets cheaply early on reduces future flexibility and increases overbooking risk. The DP captures this trade-off exactly.

Result

Maximum Expected Discounted Profit for 5-seat overbooking is $41,909.

Starting from day zero with no tickets sold, the optimal pricing policy yields an expected discounted profit of approximately $41,909 over the full booking horizon. This figure is already net of the expected overbooking penalty costs at departure.

How Prices Change Over Time

The policy derived from the DP reveals a sensible pricing pattern:

- Early in the booking horizon (many days left): the airline sets higher prices, since there is still plenty of time to sell tickets. There is no urgency to discount.
- Later in the booking horizon (few days left): prices drop to fill remaining seats and avoid flying with empty seats.
- When first-class sells out: coach demand increases, so the airline can maintain higher coach prices.
- When coach is nearly full: the algorithm naturally stops selling more coach tickets to avoid large overbooking penalties.

Task 2 - Finding the Optimal Overbooking Policy

Objective

Task 1 fixed the overbooking buffer at 5 extra seats. In Task 2 we repeat the entire DP analysis for every overbooking level from 5 to 20 extra seats, that is, maximum coach capacities from 105 to 120 and identify which level produces the highest expected profit.

$$\text{Capacity} = 100 + x, \quad \text{where } x \text{ is in } \{5, 6, 7, ..., 20\}$$
$$\text{Find: } x^* = \text{argmax } V_x(0, 0, 0)$$

Trade-Off

Allowing more overbooking creates a natural tension. Selling more tickets brings in more revenue, but if too many passengers show up, the penalty costs can exceed the extra revenue earned. The optimal overbooking level is the point where the marginal revenue from one more ticket sale is exactly offset by the expected increase in penalty costs.

Results

Running the dynamic program for all 16 overbooking levels, expected profit rises steadily from $41,909 at the 5-seat buffer, peaks at $42,171 at the 9-seat buffer, and then declines consistently through the 20-seat buffer where profit falls to approximately $41,606. The improvement from x = 5 to the optimum at x = 9 is about $262, roughly a 0.6% gain.

Optimal overbooking level: 9 extra seats (maximum capacity = 109 coach seats)

Maximum expected discounted profit: $42,171

Interpretation

Overbooking is profitable, but only up to a point. The airline's show-up probabilities (95% for coach) are high enough that selling 9 extra tickets creates a meaningful risk of exceeding capacity, which is why the optimal stops at 9 and not higher. If show-up probabilities were lower, the optimal overbooking level would be higher; if they were higher, the optimal level would be lower.

It is also worth noting that the profit differences across overbooking levels are relatively modest, less than $600 separates the best from the worst policy in this range. This means small errors in estimating show-up probabilities are unlikely to dramatically alter the conclusion that some moderate degree of overbooking is beneficial.

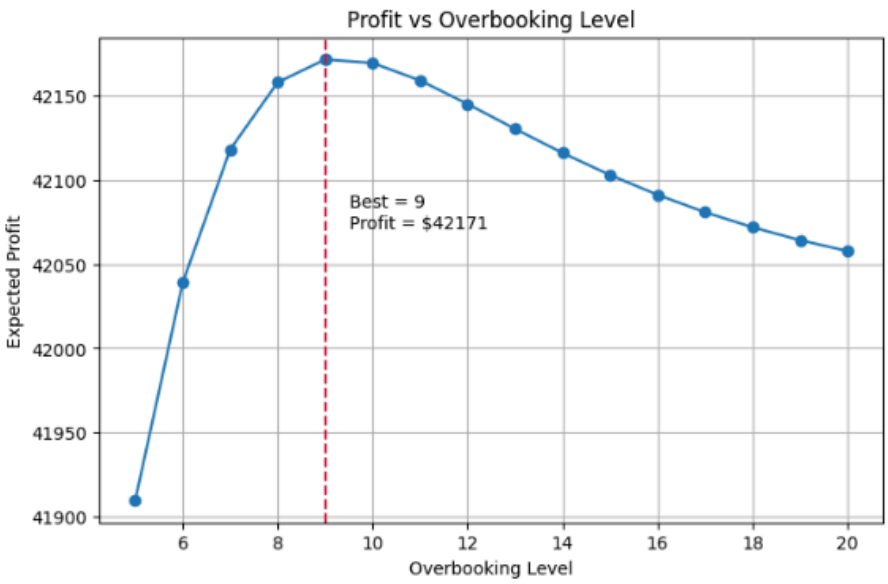

Fig 1: Profit vs Overbooking Level

The profit curve rises from x = 5 to x = 9, then falls as overbooking levels increase beyond 9. This confirms a clear optimal overbooking policy:

- Allowing 9 extra coach ticket sales (capacity 109) maximizes expected profit at approximately $42,171.
- Beyond x = 9, the rising overbooking penalties outweigh the additional revenue from extra ticket sales.
- Below x = 9, the airline leaves money on the table by being too conservative.

Profit gain from overbooking (vs. no overbooking): $42,171 - $41,909 = $262 approximately a 0.6% improvement.

TASK 3: Flexible Demand Control: The No-Sale Option

Setup

Tasks 1 and 2 gave the airline one way to manage overbooking risk: set a hard cap on coach tickets before the booking window opens, then let the DP price optimally within it. Task 3 builds on this by asking whether allowing the airline to also stop selling on certain days leads to better outcomes.

The change is straightforward. Coach pricing now includes four options every day instead of three: $300, $325, $350, or no sale. Choosing "no sale" sets coach demand to zero for that day. First-class pricing remains the same. We set the hard cap at 130 total coach tickets, giving the policy more flexibility than the Task 2 best of 109, while still avoiding too much overbooking.

The Bellman equation doesn't change. The one difference is that the DP now evaluates one additional action each day and will select "no sale" when the expected penalty from selling another ticket exceeds the revenue it would generate.

Results

Running the DP with cap = 130 and the no-sale option included, the expected discounted profit from state V(0, 0, 0) is $42,177, which is significantly higher than the Task 2 best of $42,171 at cap = 109. The flexible policy generates a higher profit than the best fixed cap from Task 2, even with a ceiling nearly 20% higher.

- Q2 Best profit (fixed cap = 109): $42,171
- Q3 profit (flexible, cap ≤ 130): $42,177
- Difference (Q3 - Q2): $6

When Does the Policy Use "No Sale"?

We tracked how often the no-sale action gets chosen across all states. Below physical capacity of 100, the DP almost never chooses it. Going past 100 is when it begins to appear more frequently, and past 120 it becomes the primary choice. The no-sale action is not a complete override, it gets triggered based on both tickets sold and days remaining, something a hard cap isn’t capable of doing.

Interpretation

The $6 increase in profit sounds miniscule, but the value of this policy is not the extra dollar value. The advantage is that the airline no longer needs to decide on a cap. In Task 2 we searched over 16 overbooking levels to find that 9 extra seats was optimal. If demand conditions shift, that search has to occur again. The Task 3 policy adjusts itself: the DP decides each day when the risk of overbooking is too high and reduces sales on its own.

The no-sale action also does not get used randomly. IBelow physical capacity, the DP keeps selling without hesitation since there is no overbooking risk yet. Once tickets sold crosses 100, no-sale starts showing up more often, and by 120 it takes over almost entirely. This is exactly the kind of state-aware behavior a fixed cap cannot replicate.

As Task 5 shows, a fixed cap becomes even less effective once demand changes seasonally. The flexible policy handles that naturally without any additional tuning, which makes it the more practical choice for a pricing department managing flights under real demand and uncertainty.

TASK 4: Sensitivity Analysis on Sale Probabilities

Setup

To test how robust the optimal policy is to errors in demand estimation, we hold ticket prices fixed at their optimal values from the Task 3 flexible policy (cap = 130) and perturb each of the five sale probabilities one at a time over a range of negative ten to positive ten percentage points. For each perturbation level we re-solve the full dynamic program and record the resulting expected profit at state V(0, 0, 0). This produces a sensitivity curve for each price point showing exactly how much the optimal profit rises or falls as the assumed demand probability moves away from its baseline.

The five probabilities tested are: coach at $300 (base: 65%), coach at $325 (base: 45%), coach at $350 (base: 30%), first-class at $425 (base: 8%), and first-class at $500 (base: 4%).

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Base expected profit (all probs at face value): $42,177.14

Profit sensitivity (central finite difference at base probabilities):
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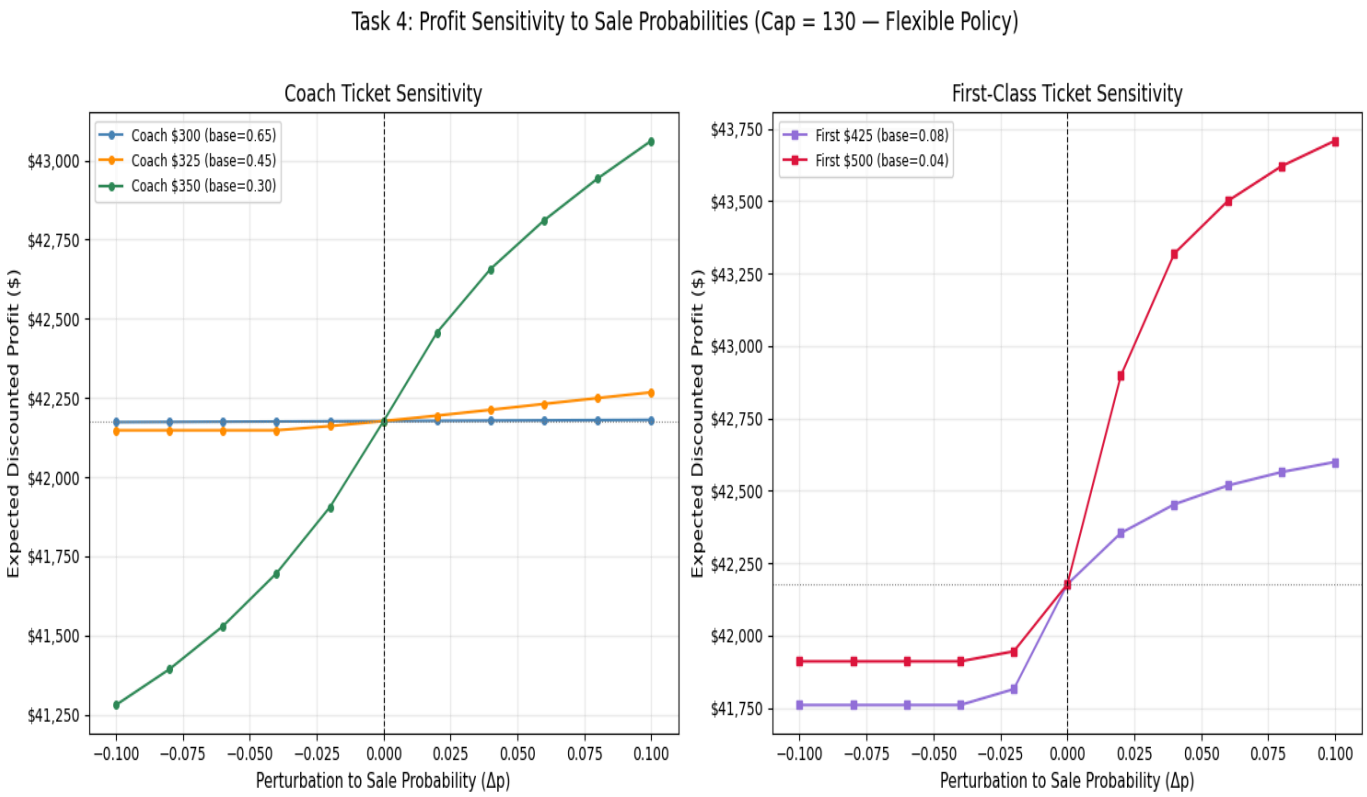
	Probability	dProfit/dProb ($/pp)
0	First $500 (base=0.04)	23807.573876
1	Coach $350 (base=0.30)	13778.838352
2	First $425 (base=0.08)	13479.825578
3	Coach $325 (base=0.45)	828.062827
4	Coach $300 (base=0.65)	36.729195



Results and Interpretation

The results are initially counterintuitive. You might expect coach tickets to dominate the sensitivity analysis since the airline sells far more of them, but the numbers tell a different story. First-class holds the top two spots in the sensitivity ranking, with the $500 ticket showing the steepest slope of all five options at a derivative of approximately $23,808 per percentage point. The $350 coach ticket comes in second at $13,779 per percentage point, and the $425 first-class ticket follows at $13,480 per percentage point. Among coach options, the $350 price point is the only one showing meaningful sensitivity, where a 10 percentage point drop in its sale probability pulls expected profit from $42,177 down to roughly $41,250, a loss of approximately $920.

The reason first-class dominates is proportional impact. The $500 ticket has a base probability of only 4%, meaning a one percentage point shift represents a 25% relative increase in demand for that ticket. Coach at $300 already operates at a 65% base probability, close to saturation, and the dynamic program has already built a policy around that reality. Shifting it by one

percentage point barely moves the needle, which is why its derivative is only $37 per percentage point.

Looking at the sensitivity curves, the slope of each line tells us how quickly profit responds to a change in that demand probability. A steep slope means the airline is highly exposed to being wrong about that price point. A flat slope means even a large estimation error barely matters. The $500 first-class and $350 coach curves diverge sharply from the baseline as you move left or right, while the $325 and $300 coach options produce lines that are nearly flat across the entire range tested. For those two price points, the airline could be off by the full ten percentage points in either direction and barely notice it in the bottom line.

Importantly, none of the sensitivity curves drop sharply within the five percentage point range closest to the baseline. Profit degrades gradually in both directions for every price point, which means the optimal policy would not collapse from a minor forecasting mistake. The area of greatest concern remains the first-class premium price point: first-class demand is inherently harder to forecast given the small number of tickets sold, and the financial consequences of underestimating it are the largest of any price option available to the airline.

TASK 5:

In the previous tasks, we assumed that customer demand remained constant throughout the 365-day booking horizon. However, real-world airline data suggests that demand is often "seasonal," with the probability of a sale fluctuating as the departure date approaches. Task 5 relaxes the constant demand assumption by introducing a time-dependent seasonality multiplier.

Mathematical formulation

To model this shift, we define a seasonality multiplier, $M(t)$, where t represents the number of days remaining until the flight departs:

$$M(t) = 0.75 + t/730$$

This multiplier scales the baseline sale probabilities for both coach and first-class tickets. For example, on day t=100, the multiplier is approximately 0.887, reducing a baseline coach sale probability of 65% to roughly 57.65%. Conversely, very early in the booking horizon (e.g., t=365), the multiplier is 1.25, increasing the likelihood of a sale.

The Bellman equation is updated to incorporate this daily fluctuation:

$$V(t, c, f) = \max_{p_c, p_f} \mathbb{E}_t \left[\text{Revenue} + \delta \cdot V(t+1, c', f')\right]$$

where the expectation is now specific to the adjusted probabilities for day t.

Methodology

We re-ran the backward induction algorithm, recalculating the sale probabilities for every one of the 365 days. Because the underlying demand structure changed, we also performed a new optimization search across overbooking levels (5 to 15 extra seats) to determine if the 9-seat buffer identified in Task 2 remained the most profitable choice under seasonal conditions.

Results

The optimal overbooking level for the seasonality model remains 9 extra seats (maximum capacity of 109 coach tickets).

Overbooking Level	Expected Discounted Profit
Overbooking 7	$41,804.96
Overbooking 8	$41,844.15
Overbooking 9 (Optimal)	**$41,857.70**
Overbooking 10	$41,855.91
Overbooking 11	$41,845.99

The maximum expected profit of $41,857.70 is lower than the Task 2 static demand profit of $42,171.35. This decrease occurs because the seasonality multiplier is below 1.0 for more than half of the booking horizon ($t < 182.5$), leading to a lower total volume of ticket sales over the year.

Despite the lower overall demand, the 9-seat peak remains remarkably stable. This suggests that the trade-off between marginal ticket revenue and the expected $425 penalty for denied boarding is robust to linear shifts in demand volume.

While the fixed cap remains 9, a more sophisticated dynamic booking limit would allow the airline to vary its risk tolerance by month. As demand "cools off" closer to the flight (where the multiplier drops toward 0.75), the airline could technically afford a higher overbooking buffer because the probability of actually filling those extra seats is lower than it was during the high-demand early phase.

Conclusion

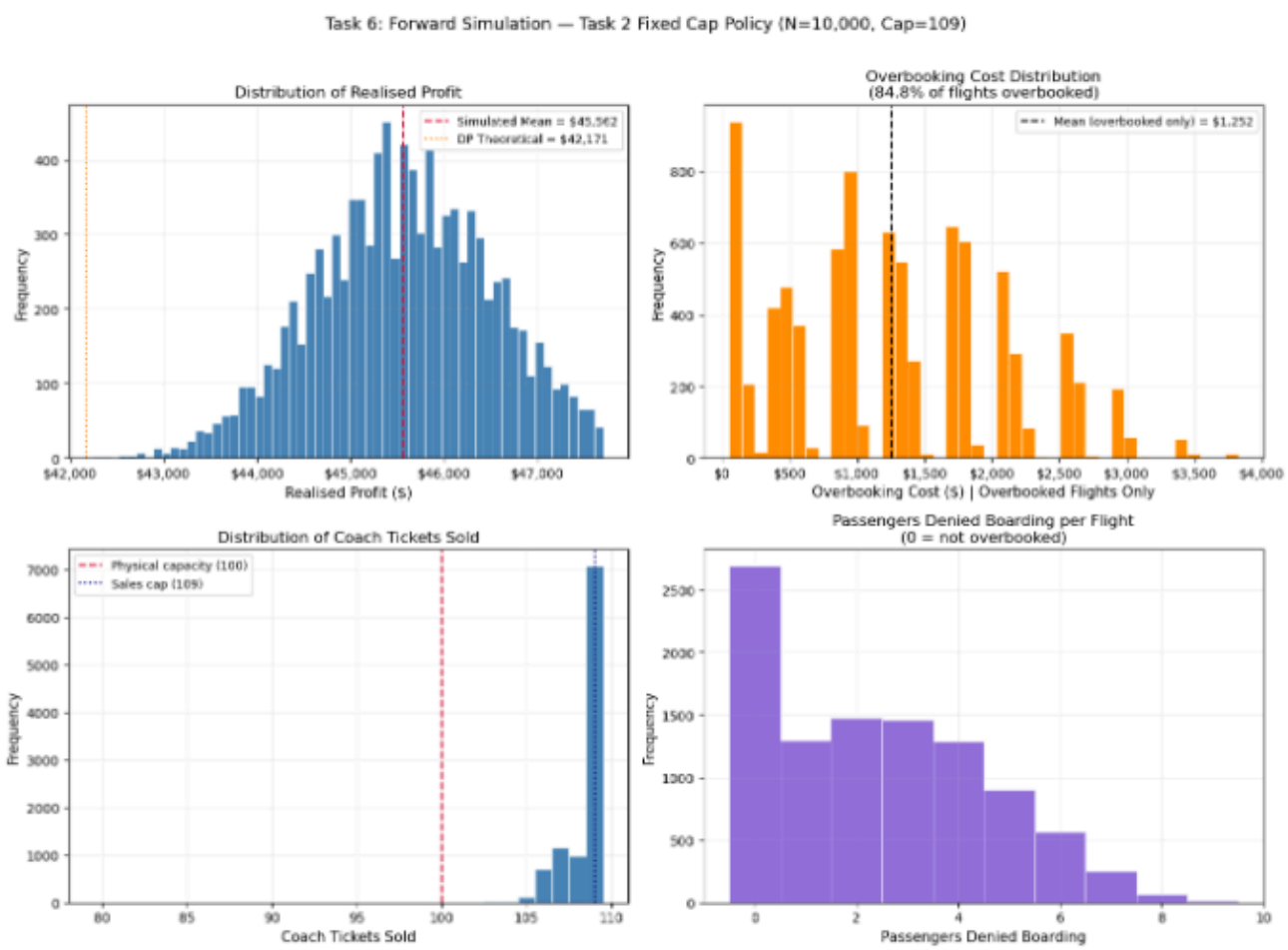
Seasonality confirms that a "one-size-fits-all" overbooking cap is a simplification. While 9 seats is the best single number for the entire year, the airline would maximize profit further by adopting the Flexible No-Sale Policy from Task 3, which allows the pricing department to stop selling tickets based on the exact risk profile of the day, effectively automating a seasonal response to demand.

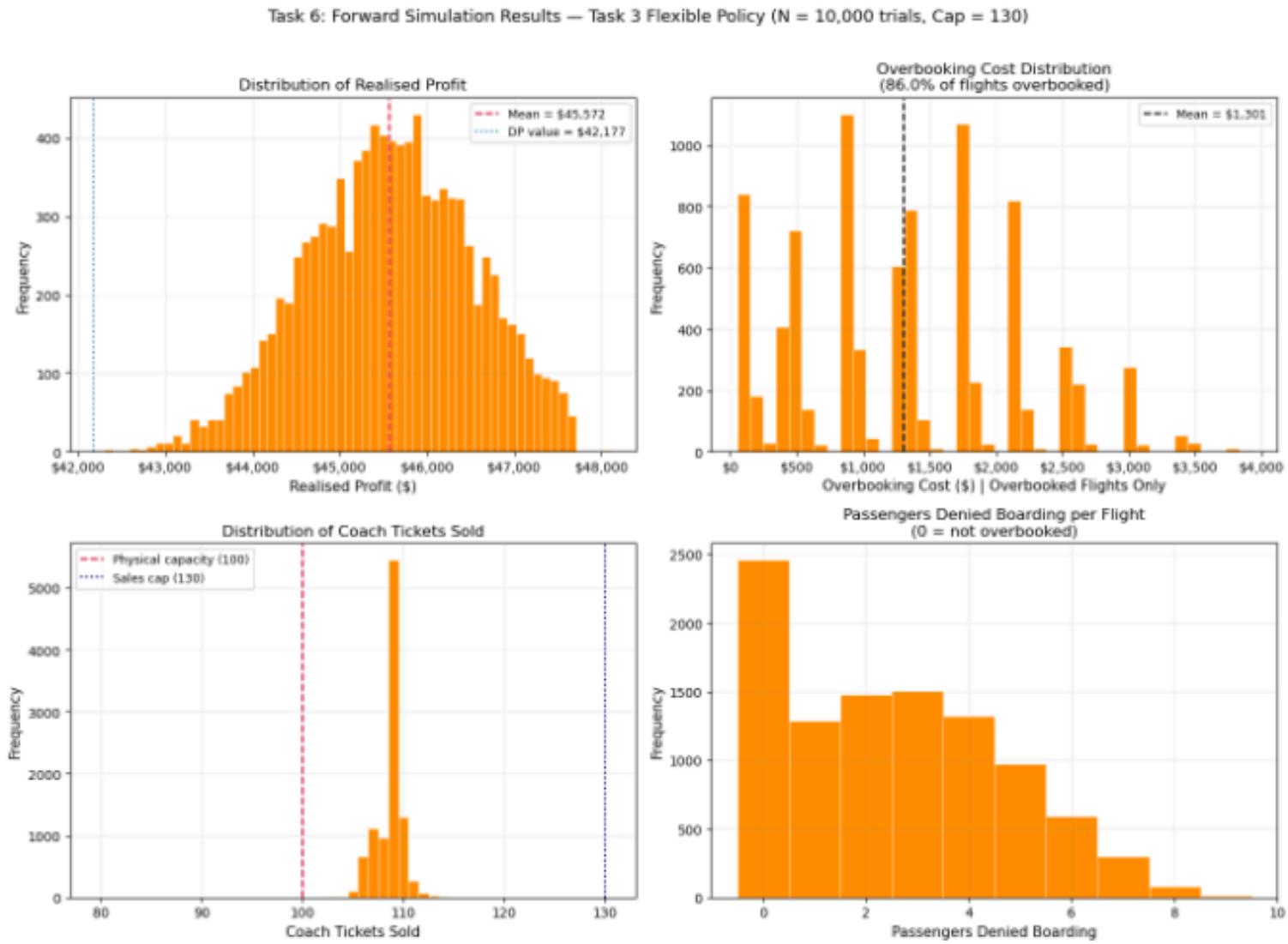
TASK 6: Forward Simulation of the Optimal Policy

Setup

Tasks 1 through 5 solve the dynamic program backwards to derive the optimal pricing policy. Task 6 evaluates that policy by simulating it forward 10,000 times, drawing actual random demand outcomes at each of the 365 selling days and recording what happens at departure when passengers show up. We simulate both the Task 2 fixed-cap policy (cap: 109) and the Task 3 flexible no-sale policy (cap: 130) under the same conditions using the same random seed, so results are directly comparable.

Each trial begins at state (t = 0, c = 0, f = 0), follows the optimal policy each day to draw Bernoulli sales, advances the state, and at day 365 draws binomial show-up counts to realize the overbooking cost. We record total undiscounted ticket revenue, overbooking cost paid, passengers bumped to first class, passengers denied boarding, and total profit for every trial.





Policy Cap 109: Task 2 Fixed Cap Results

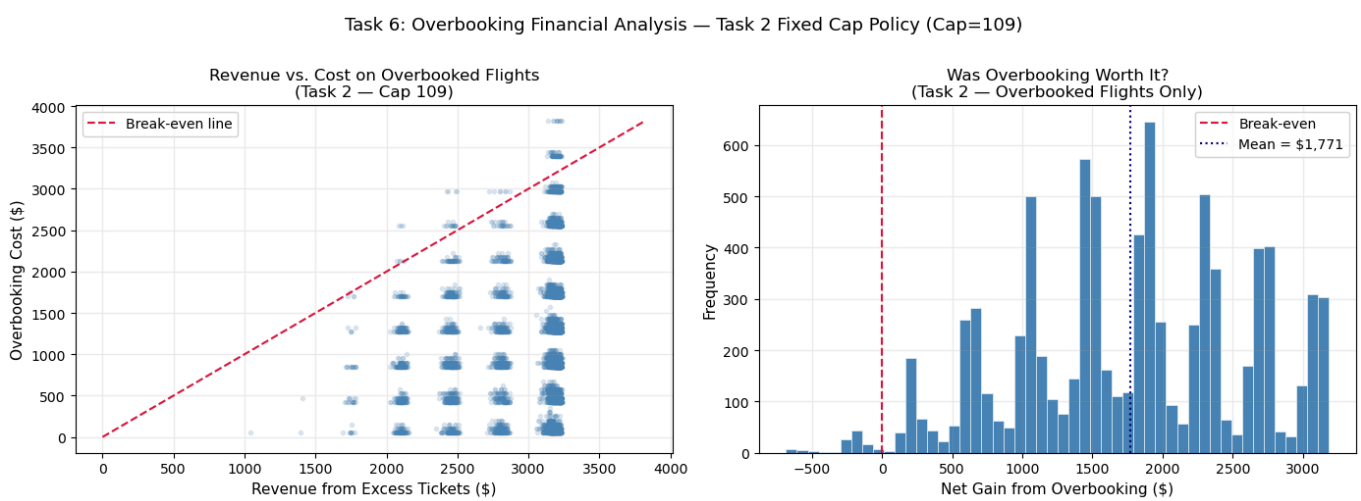
The simulated mean profit of $45,562 exceeds the dynamic programming theoretical value of $42,171 by $3,391. This gap exists because the dynamic program discounts all future revenue

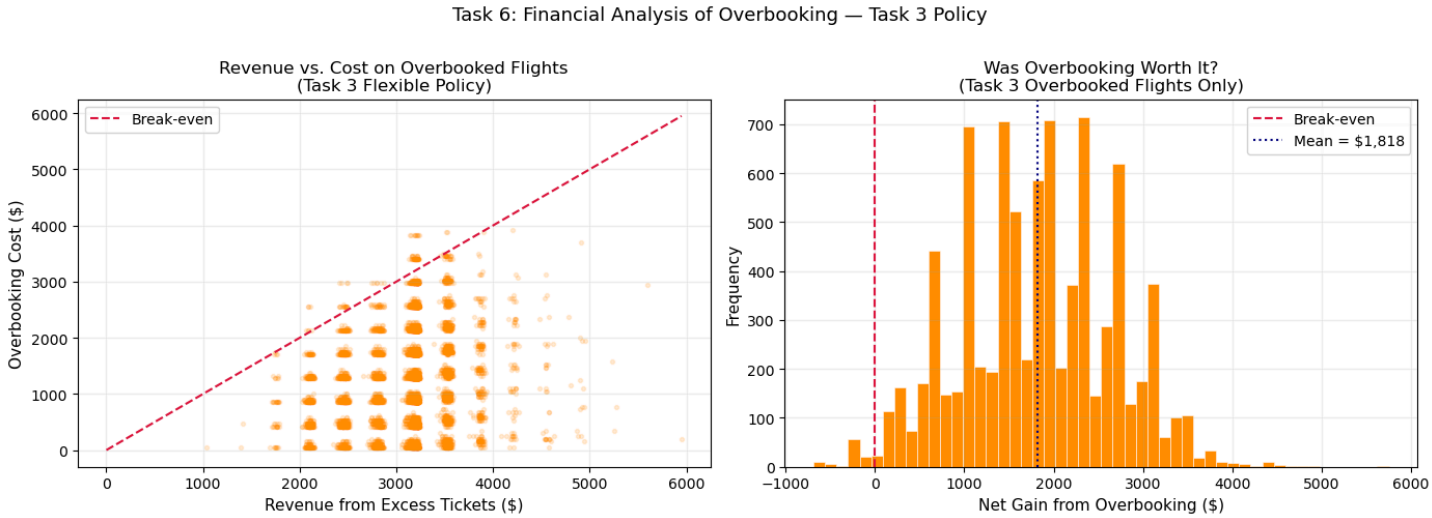
using the daily discount factor, while the simulation records raw undiscounted ticket revenue. The DP value is a conservative lower bound on cash collected, not a ceiling. The overbooking rate came in at 84.75% of flights, costing an average of $1,061 per flight across all trials. When isolating only the flights that were actually overbooked, the average cost spikes to $1,252, reflecting that the distribution of overbooking costs is right-skewed: most overbooked flights incur modest costs, but occasional high-overflow events pull the conditional mean up. The coach tickets sold distribution shows the policy nearly always sells right up to the hard cap of 109, confirming that the dynamic program consistently finds value in selling all allowable tickets. When passengers are denied boarding, 2 to 3 people is the most likely outcome. Profit volatility across the 10,000 trials is $940.

Policy Cap 130: Task 3 Flexible No-Sale Results

The Task 3 flexible policy produces a simulated mean profit of $45,572, just $10 higher than the Task 2 policy, while the dynamic programming theoretical value of $42,177 is also marginally better. The overbooking rate is slightly higher at 85.96%, but the average overbooking cost is lower at $1,118. This seems counterintuitive at first: a wider cap should allow more overbooking, yet costs are lower. The explanation is that the no-sale option lets the policy cut off coach sales dynamically on days when the overbooking risk becomes too high, preventing the worst-case ticket accumulations that would otherwise occur near the cap. Profit volatility at $952 is nearly identical to the Task 2 result, confirming that neither policy carries meaningfully more financial risk than the other. The coach tickets sold distribution reflects the flexibility of this policy: unlike Task 2 where tickets almost always hit 109, the Task 3 distribution shows a wider spread, with the no-sale action holding back sales on high-risk days.

Is Overbooking Financially Justified?





To directly answer whether overbooking is good or bad for revenue, we isolated the trials where overbooking actually occurred and compared the revenue collected from the extra tickets sold above physical capacity against the overbooking penalties paid. For the Task 2 policy, the average net gain from overbooking on overbooked flights is $1,912. For Task 3 it is $1,940. Across both policies, approximately 98% of flights that are overbooked still result in a net positive transaction for the airline after paying all bumping and denied boarding costs. The scatter plots of revenue versus overbooking cost confirm this visually: the vast majority of overbooked flights fall below the break-even line, meaning the revenue from extra seats exceeds the penalty paid. Overbooking is not a necessary evil that the airline tolerates for capacity reasons. It is an active profit driver, and removing it would cost the airline thousands of dollars in annual revenue under either policy.